Reclassification footnote (2026-05-26 EOD addendum, binding 2026-05-26) This paper is the addendum (-AddA) to the parent `Paper-07-ext: SO(10) anomaly cancellation via the spinor-trace method`, written in response to the 2026-05-26 C1/C2/C3 epistemic-category reclassification (Math411-AddB §10 operator binding). Canonical TECT positioning (binding 2026-05-26): *TECT is an emergent vacuum-and-gravity framework (7 C1 pillars at T6/T7) with an unresolved gauge-sector completion (Pillar 6 standard routes EXHAUSTED) and phenomenological dark-matter/quantum-origin pillars open.* TOE-level claim SUSPENDED until non-standard rescue independently derived.

I. BACKGROUND AND PARENT-PAPER RECAP

The parent paper Paper-07-ext (drafted in Wave 4 of the Papers Track) presented the Math157 SO(10) spinor-trace anomaly cancellation as a quantum-consistency gate closure (GAP-3). The argument is group-theoretic: the six anomaly coefficients $\mathrm{Tr}(T^a\{T^b, T^c\})$ for the SM subgroups (cubic SU(3), cubic SU(2), mixed $U(1)$-SU(3)^2, mixed $U(1)$-SU(2)^2, gravitational, cubic $U(1)$) are computed as traces on the **16** representation under the standard $SO(10) \supset SU(5) \times U(1)_\chi \supset G_{\rm SM} \times U(1)_\chi$ embedding chain. All six evaluate to exact zero. This is verified numerically in `Codes/supplementary/Math157_anomaly_trace_verification.py`.

II. MATHEMATICAL CONTENT SURVIVAL

The Math157 result is a *purely group-theoretic* statement. It assumes only: **(H1)** the SO(10) gauge group structure is realised; **(H2)** the chiral fermion content sits in the **16** representation; **(H3)** the embedding chain $SO(10) \supset SU(5) \times U(1)_\chi \supset G_{\rm SM} \times U(1)_\chi$ is standard. The 2026-05-26 reclassification touches *none of these hypotheses.* The reclassification refutes the standard-framework dynamical-realisation *routes* for SO(10) emergence (Pathway A Math157-revival, Pathway B SO(4)+SO(4), T_{2u} doubling, Stueckelberg-direct), but it does not refute the SO(10) anomaly-cancellation lemma itself. The lemma stands as a conditional theorem: "GIVEN H1+H2+H3, anomalies cancel."

III. RECLASSIFICATION IMPLICATIONS

Post-reclassification, the proper classification of the Math157 result is:

- **Mathematical content tier**: T7 PROVED (group-theoretic, no dynamical assumption beyond H1+H2+H3 above).
- **Pillar attribution**: contributes to Pillar 6 (gauge group emergence) as a downstream consistency lemma; does NOT contribute to Pillar 6 promotion.
- **Conditional structure**: "T6 PROVED CONDITIONAL on Pillar 6 emergence at SO(10) level by ANY mechanism."
- **Standalone publishability**: the result is publishable as a standalone group-theory contribution (e.g., as a Note in Physical Review D or a Brief Report); the standalone publication does NOT depend on TECT's Pillar 6 dynamical-realisation status.

This is the canonical handling of an upstream-conditional result whose downstream hypothesis (Pillar 6 dynamical realisation in standard framework) has been refuted: the result is preserved as a conditional lemma, not retracted.

IV. FORWARD PATH: NON-STANDARD RESCUES

If a non-standard Pillar 6 rescue (Math411-AddB-AddA fermion-condensate aux channels, Math411-AddB-AddB holographic / dim-reduction emergence) is independently derived *and* produces the SO(10) gauge structure with **16** matter content, the Math157 lemma immediately applies and the SM anomaly cancellation is automatic. Until such a rescue is derived, the Math157 result remains a conditional lemma in the GUT-theory library and contributes to the C1 core-emergence sector only as a consistency lemma.

V. CROSS-REFERENCES

- Math157 (canonical anomaly-cancellation note)
- Paper-07-ext (parent paper)
- Math411-AddB §10 (canonical 2026-05-26 reclassification source)
- Codes/supplementary/Math157_anomaly_trace_verification.py (numerical verification)
- Docs/status/TOE-FACT-SHEET.md (canonical scoreboard)